Introduction to Time Series Forecasting

Practical Methods and Applications

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Ancona Erasmus Lecture

Outline

1. Introduction to Forecasting
2. Naive Forecasting Methods
3. Time Series Regression Models
4. Forecast Intervals and Uncertainty
5. Forecast Evaluation

References

- Hyndman and Athanasopoulos (2021), URL: https://otexts.com/fpp3/
  - Gretl Replication Files: https://github.com/gretl-project/forecasting_principles
- Castel et al. (2019)
- Diebold (2024)
- Elliott and Timmermann (2016)
- Enders (2015)

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Introduction to Forecasting

"A forecast is any statement about the future."

Castel et al. (2019)

What is Forecasting and Why Does It Matter?

Forecast

Seeks to predict the future as accurately as possible, taking into account all available information, including historical data and knowledge of future events that may influence forecasts. Hyndman and Athanasopoulos (2021)

- Forecasting is integral to decision-making in business and economics.
- **Business:** supports operations, staffing, finance, and strategy.
- **Economics:** informs GDP, inflation, unemployment, and policy planning.
- Forecast horizons depend on purpose:
  - **Short-term**: operational control
  - **Medium-term**: budgeting and capacity planning
  - **Long-term**: investment and strategic/policy direction

Predictability: What Determines It?

1. How well we understand the factors that contribute;
2. How much data is available;
3. How similar the future is to the past (structural stability);
4. Whether forecasts can influence what we are trying to predict (self-fulfilling prophecies).

Short-term Electricity Demand Forecasts...

...can be forecast accurately because all **four predictability conditions** are usually met:

- ✓ Good understanding of contributing factors: Temperature, with minor effects from calendar deviations like holidays and economic conditions.
- ✓ Long history: Multiple years of electricity demand data are available, along with decades of weather data.
- ✓ Structural stability: For short-term forecasts, we can assume stable demand behavior.
- ✓ No feedback effects: Demand forecasts have little or no influence on consumer behavior.

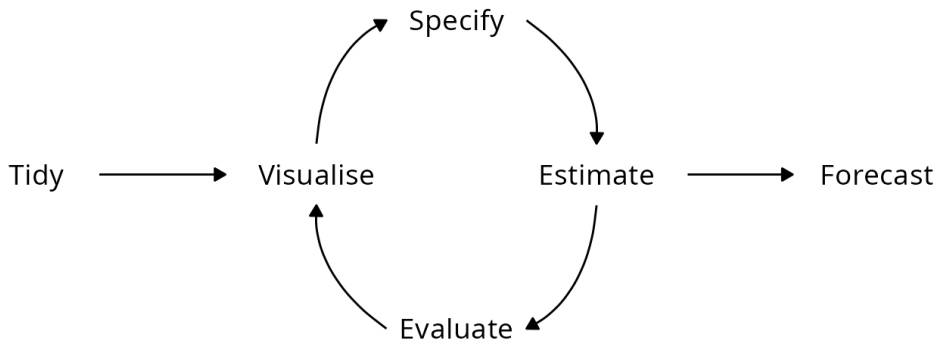
Exchange Rate Forecasts Are More Difficult

- ✓ Plenty of available data.
- × Deep understanding of factors influencing exchange rates.
- × The future mimics the past only to a high extent.
- × Public forecasts have no effect on the exchange rate.

Result: Exchange rate forecasts are often inaccurate. This is consistent with the **random walk hypothesis** in finance, which states that exchange rates follow a random walk and are therefore unpredictable.

The Forecasting Workflow

1. **Collect & Clean**: Collect and preprocess data
2. **Exploratory Analysis**: Understand data patterns
3. **Model Selection**: Choose appropriate method
4. **Estimation**: Fit model to training data
5. **Evaluation**: Assess accuracy on test data
6. **Forecasting**: Generate predictions



Key Concepts

- **Point Forecast**: Single predicted value
  - Example: "GDP growth next quarter = 2.5%"
- **Prediction Interval**: Range of plausible values
  - Example: "GDP growth 95% CI: [1.8%, 3.2%]"
- **Probabilistic Forecast**: Full distribution of future values
  - Example: "GDP growth has 30% chance of being above 2%"
- **Training Data**: Used to fit the model
- **Test Data**: Held out to evaluate forecast accuracy
- **In-sample vs. Out-of-sample**:
  - In-sample fit does NOT guarantee good forecasts (overfitting)
- **Naive/simple vs. Sophisticated Methods**:
  - Always compare to simple benchmarks (e.g., mean, last value)
  - Sophisticated methods must outperform naive ones to be useful
  - Simple methods are often hard to beat, especially for short-term forecasts

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Naive Forecasting Methods

Why Start with Naive Methods?

- **Benchmark**: Any sophisticated method should beat naive forecasts or simple forecasting methods
- **Interpretability**: Easy to understand and explain
- **Practical**: Often surprisingly effective — at least as useful benchmarks
- **Big-Data**: Applying naive methods to *zillions* of time series (e.g., retail sales by product) can be done in reasonable time and often even by SQL queries.

Read also Rob J Hyndman's blog post on the relevance of benchmarks: https://robjhyndman.com/hyndsight/benchmarks

Mean Method

Forecast all future values as the historical average:

$$\hat{y}_{T+h|T} = \bar{y} = \frac{1}{T}\sum_{t=1}^{T} y_t \tag{1}$$

for historical data $y_1, y_2, \ldots, y_T$.

When to use: Stationary data without trend, seasonality and minor persistence.

Robust version: Median method (forecast = median of historical data) is more robust to outliers.

Naïve Method (Random Walk)

Forecast: the last observed value

$$\hat{y}_{T+h|T} = y_T \tag{2}$$

When to use: Random walk data (most economic/financial series)

- Stock prices, exchange rates, unit root processes
- Adapts to level shifts and trends automatically

Random Walk plus Drift Method

Forecast: last value plus average historical change

$$\hat{y}_{T+h|T} = y_T + h \underbrace{\left(\frac{y_T - y_1}{T - 1} \right)}_{\text{average change}} \tag{3}$$

Interpretation: Extrapolate linear trend from first to last observation

When to use: Data with steady trend (GDP, population, cumulative sales)

Seasonal Naïve Method

Forecast: last observed value from same season

$$\hat{y}_{T+h|T} = y_{T+h-m(k+1)} \tag{4}$$

where m = seasonal period (e.g., 4 for quarterly, 12 for monthly), and k is the integer part of $(h-1)/m$ (i.e., the number of complete years in the forecast period prior to time $T+h$).

Example: For quarterly data, forecast Q2 2024 = Q2 2023 value

When to use: Highly seasonal data without trend (retail sales, tourism, energy)

Seasonal Mean Method

Idea: Forecast using the historical mean of the corresponding season.

$$\hat{y}_{T+h|T} = \bar{y}_{s(T+h)} \quad \text{with} \quad \bar{y}_j = \frac{1}{N_j} \sum_{t \in \mathcal{S}_j} y_t \tag{5}$$

- **Weekday frequency:** one mean for each weekday
- **Monthly frequency:** one mean for each month (12 means)
- **Quarterly frequency:** one mean for each quarter (4 means)

When useful: Stable seasonal pattern, weak trend.

Comparing the Methods

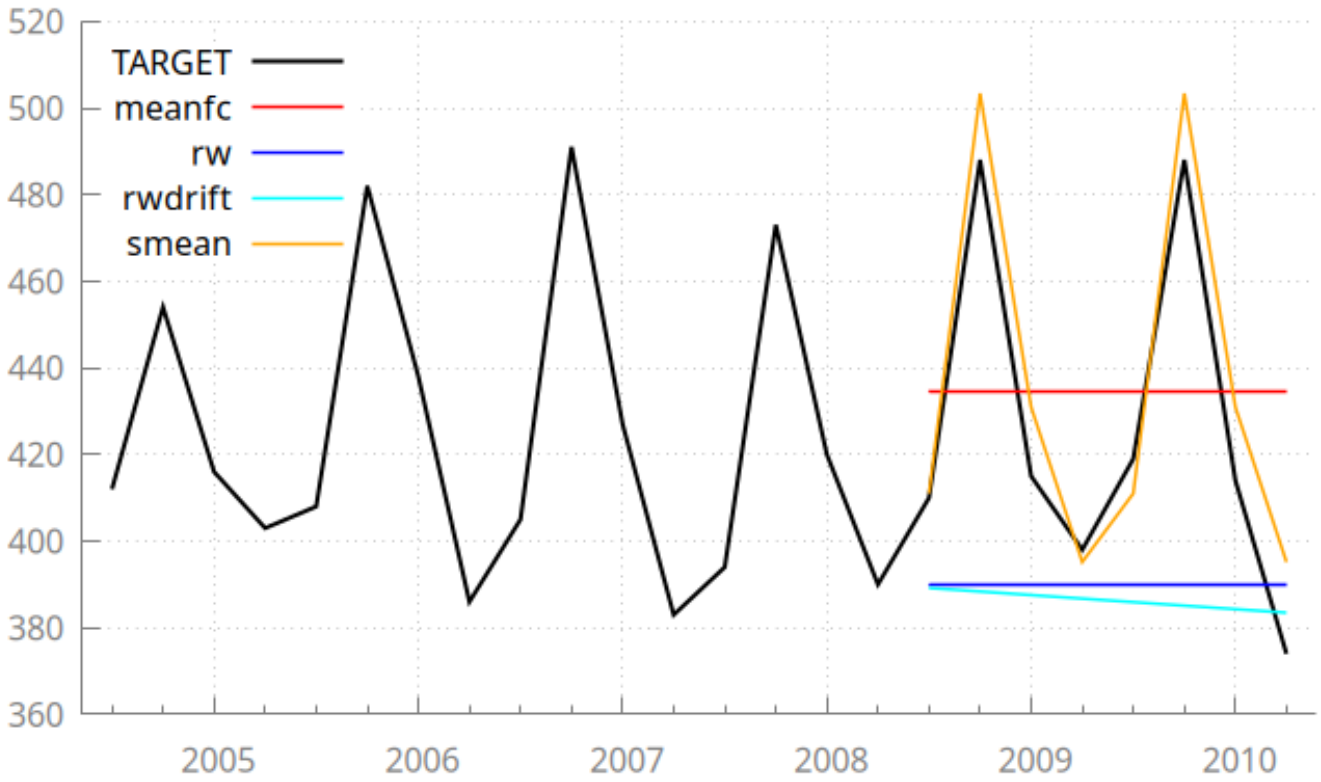

Figure 1: Australian quarterly beer production: Mean, Random Walk, Random Walk + Drift and Seasonal Naïve

When Do Naive Methods Fail?

- Data with **autocorrelation**: Past values contain predictive info
- **Complex patterns**: Trends, level shifts, cycles
- **Structural dependencies**: Future depends systematically on past

More advanced time-series models:

- (S)ARIMA(X)
- Exponential Smoothing
- Regression with ARMA errors (ARIMAX)
- ARDL and VAR
- Structural Time Series Models (e.g., unobserved components)
- Etc.

Naive/simple Methods Using Gretl

Gretl provides all relevant functions for computing naive methods.

Make use of the ‘arima‘ and ‘ols‘ commands.[1]

The “naiveFC” package provides a convenient interface for generating forecasts from naive methods, including point forecasts and prediction intervals:

- Doc: https://gretl.sourceforge.net/current_fnfiles/unzipped/naiveFC.pdf
- Github: https://github.com/atecon/naiveFC/tree/master/
- Package needs some update, but works

[1] See relevant chapters in the User’s Guide. URL: https://gretl.sourceforge.net/gretl-help/gretl-guide.pdf

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Time Series Regression Models

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Some useful predictors

Trend Regression

Definition: Model linear trends in time series data

Model:

$$y_t = \beta_0 + \beta_1 \cdot t + \varepsilon_t \quad t = 1, 2, \ldots, T \tag{6}$$

- β_0: Starting level
- β_1: Average change per time period
- **When to use:** Trend stationary series with deterministic trend

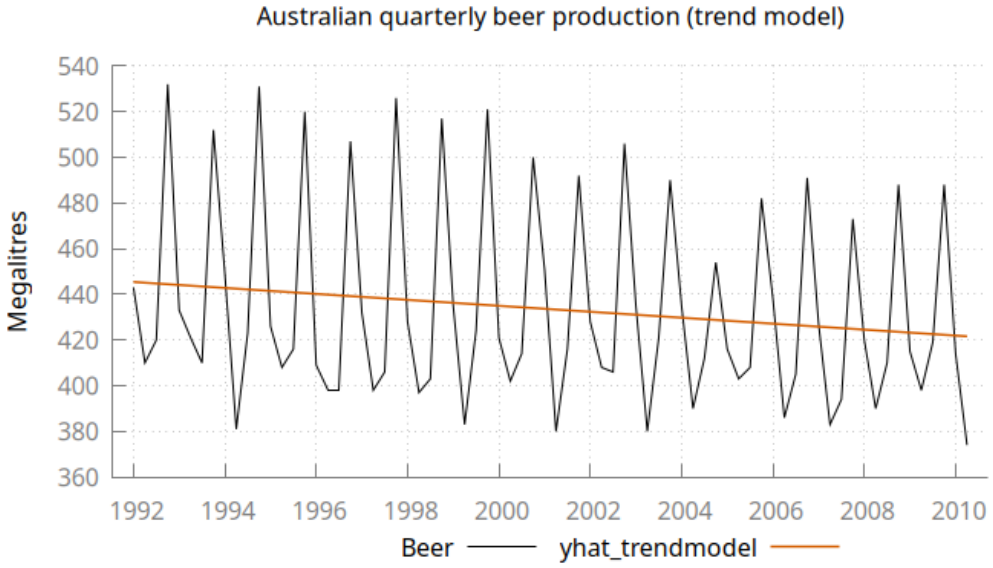

Figure 2: In-sample fit of trend regression model to Australian beer production data

Dummy Variables for Categorical Effects

Encode categorical information into regression models.

Common uses:

- **Holidays/Special events**: Public holidays[2], Sport events
- **Intervention effects**: Marketing campaigns, policy changes
- **Outliers**: Handle special one-time events

Interpretation: Each coefficient represents the effect **relative to the reference (omitted) category**

[2] See the gretl package "holidays" for built-in holiday calendars: https://gretl.sourceforge.net/current_fnfiles/holidays.gfn.

Seasonal Dummy Variables

Example: **Quarterly** with 4 categories $\Rightarrow$ use 3 dummy variables

Quarter	$d_{1,t}$	$d_{2,t}$	$d_{3,t}$
Q1	1	0	0
Q2	0	1	0
Q3	0	0	1
Q4	0	0	0

Extensions – handle different seasonalities[3]

- **Weekly data:** 6 dummies (for Mon, ..., Sat; Sun is reference)
- **Monthly data:** 11 dummies (for Jan–Nov; Dec is reference)

[3] Fourier series (sine and cosine terms) provide an alternative for modelling seasonality, especially useful for long seasonal periods. See https://otexts.com/fpp3/dhr.html for details.

Seasonal Model plus linear Trend

Model example: (quarterly frequency)

$$y_t = \beta_0 + \beta_1 t + \beta_2 d_{2,t} + \beta_3 d_{3,t} + \beta_4 d_{4,t} + \varepsilon_t$$

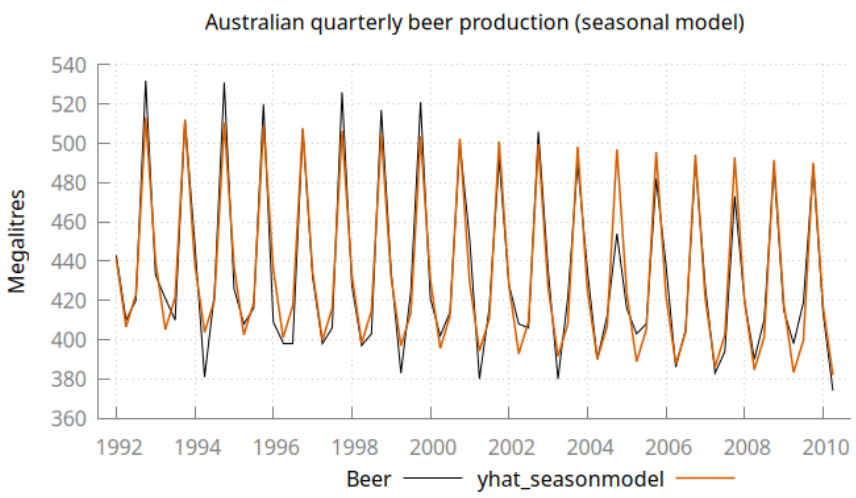

Figure 3: In-sample fit of seasonal regression model

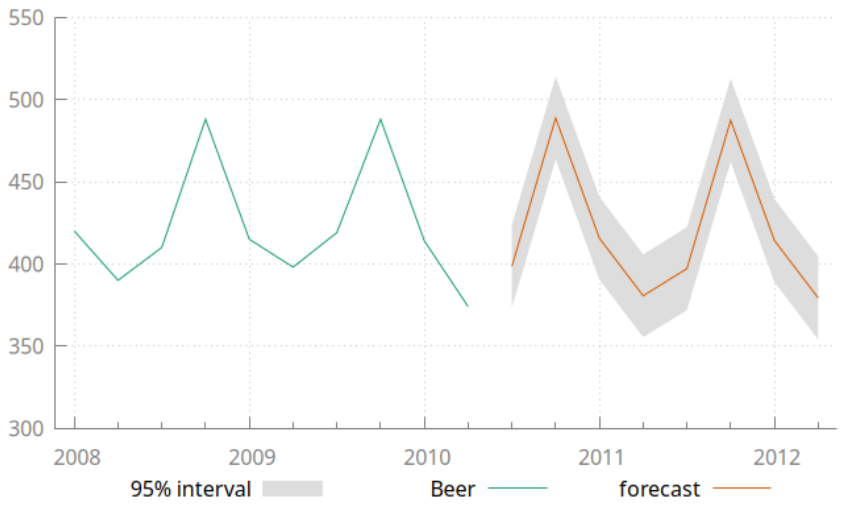

Figure 4: Out-of-sample forecast with prediction intervals

Distributed Lags: Modeling Lagged Effects

Motivation: Some predictors have effects lasting multiple time periods

Example: Advertising campaign

- Sales impact occurs not only in the campaign month, but also in subsequent months $\rightarrow$ Need to capture cumulative/delayed effects

Distributed lag model:

$$y_t = \beta_0 + \beta_1 X_{t-1} + \beta_2 X_{t-2} + \cdots + \beta_m X_{t-m} + \varepsilon_t$$

where X_{t-j} is the predictor value j periods in the past.

Forecasting Challenge:

- To forecast y_{T+h}, we need values $X_{T+h-1}, X_{T+h-2}, \ldots, X_{T+h-m}$
- Many of these future X values are **unknown**
- Must either: (1) use assumptions about future X, or (2) forecast X as well
- Makes out-of-sample forecasting more difficult and uncertain

Distributed Lags: Example with Beer and Cement

Model example:

$$Beer_t = \beta_0 + \beta_1 time + \beta_2 \text{Cement}_{t-6} + \beta_3 \text{Cement}_{t-7} + \beta_4 \text{Cement}_{t-8} + \varepsilon_t$$

(You may also add a seasonal component if needed, but we focus on the distributed lag aspect here)

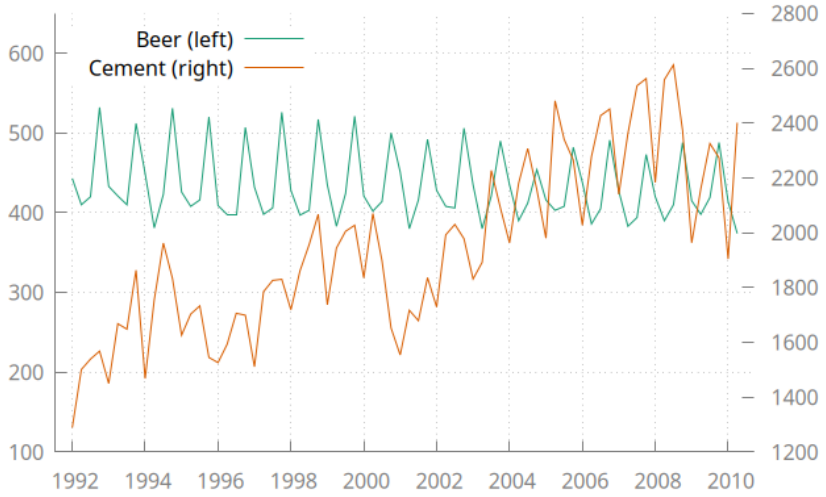

Figure 5: Historical data for beer production and cement production

Figure 6: Out-of-sample forecast with prediction intervals

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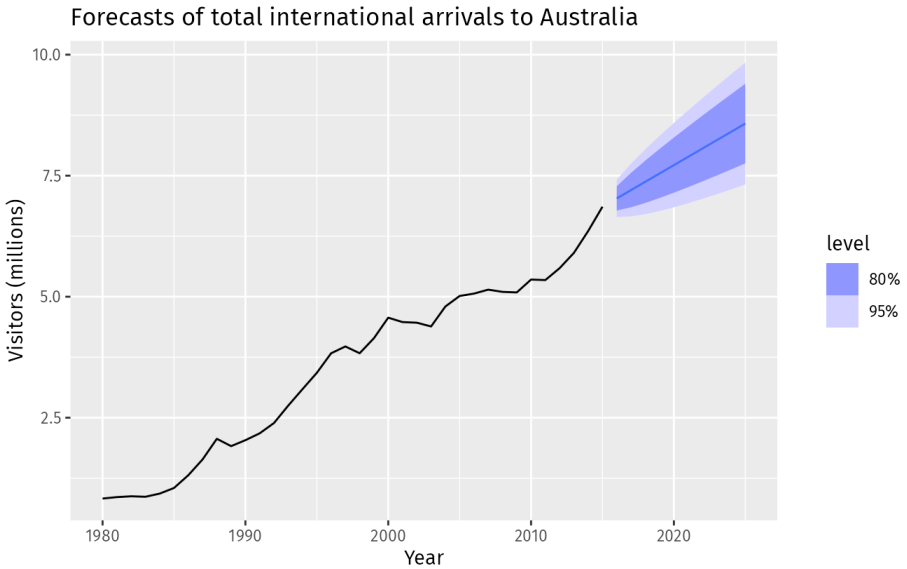
Forecast Intervals and Uncertainty

Why Prediction Intervals Matter

- Point forecasts alone are incomplete
- Decision-makers need to know the **range of uncertainty**
- Narrow intervals ⇒ more confident forecast
- Wide intervals ⇒ high uncertainty

Example:

- Bad: "Sales will be $100M"
- Better: "Sales will be $100M [95% CI: $80M–$120M]"



Interpretation of Prediction Intervals

80% prediction interval: If we repeat this forecast many times, the interval will cover actual values in about 80% of the time.

95% prediction interval: Higher coverage (wider band) for more confidence. Less useful for decision-making if too wide.

Methods to construct intervals:

- **Asymptotic**: Based on normality of residuals
- **Bootstrap**: Resampling from residuals (or forecast errors) (more robust)

Note: Software packages typically provide interval estimates for classical models (e.g., ARIMA forecasts)

Benchmark Methods: Forecast Standard Deviations

Mathematical foundation: Derive h-step forecast uncertainty under uncorrelated residuals

Let $\hat{\sigma}_h$ = standard deviation of h-step forecast distribution, and $\hat{\sigma}$ = residual standard deviation

Benchmark Method	**h-step Forecast Std Dev**
Mean	$\hat{\sigma}_h = \hat{\sigma}\sqrt{1 + 1/T}$
Naïve	$\hat{\sigma}_h = \hat{\sigma}\sqrt{h}$
Seasonal Naïve	$\hat{\sigma}_h = \hat{\sigma}\sqrt{k+1}$
Drift	$\hat{\sigma}_h = \hat{\sigma}\sqrt{h\left(1 + \frac{h}{T-1}\right)}$

Source: https://otexts.com/fpp3/prediction-intervals.html

Alternatively: use bootstrap methods (no need to rely on distributional assumptions).

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Forecast Evaluation

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Of Point Forecasts

Motivation

- Performance on known data is not a good indicator of prediction performance on unknown data!
- **Overfitting**: A model that adapts too strongly to the training data.
- A useful model should be able to *generalize*.
- **Generalization**: The ability of a model to predict on unknown data.
- **Cross-validation**: A method for estimating the generalization performance of a model.
- Evaluating *out-of-sample* prediction performance provides a measure of model quality.

Training and Test Errors

We distinguish between:

1. **Training error**: Residuals aka *in-sample error*
2. **Test (Generalization) error**: Forecast error when predicting unseen test data (*out-of-sample* prediction).

Goal: Minimize some loss function based on test error.

The training error can dramatically underestimate the test error.

For model selection and evaluation, we evaluate the test error.

Train-Test Split

Why separate data?

- Training data: Fit the model, estimate parameters
- Test data: Evaluate forecast accuracy on new data

Typical split: 80% training, 20% test

- Depends on: Data length, forecast horizon
- Rule: Test set $\geq$ longest forecast horizon

Figure 8: Train-test split for time series forecasting

Important: Maintain temporal order (no random shuffling!)

Forecast Errors

Definition: Difference between observed and forecast value

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$$

Key distinction:

- **Residuals**: Errors on training data (one-step ahead)
- **Forecast errors**: Errors on test data (can be multi-step)

Why it matters: Residuals underestimate true forecast uncertainty

Example: Calculating Forecast Errors & Metrics

	Actual	Forecast	Error	Percent. error
2007 Q1	427	423.69	3.31	0.78
2007 Q2	383	386.88	−3.88	−1.01
2007 Q3	394	404.71	−10.71	−2.72
2007 Q4	473	483.59	−10.59	−2.24
2008 Q1	420	423.81	−3.81	−0.91
2008 Q2	390	385.42	4.58	1.17
2008 Q3	410	403.25	6.75	1.65
2008 Q4	488	482.13	5.87	1.20
2009 Q1	415	422.35	−7.35	−1.77
2009 Q2	398	383.96	14.04	3.53
2009 Q3	419	401.79	17.21	4.11
2009 Q4	488	480.67	7.33	1.50
2010 Q1	414	420.89	−6.89	−1.66
2010 Q2	374	382.50	−8.50	−2.27
MAE			7.92	
RMSE			8.82	
MAPE				1.89%
MASE			0.54	

Table 1: *Accuracy measures computed from ARIMA forecasts for the 14 observations in the test dat*

Figure 9: 14-h-step forecasts for *one* training period. Source: Hyndman (2014)

Mean Error (ME) I: Forecast Bias

Definition: Average forecast error

$$\text{ME} = \text{mean}(e_t) \qquad \text{where} \quad e_t = y_t - \hat{y}_t \tag{7}$$

Interpretation (bias):

- ME > 0: systematic **under-prediction** (forecasts too low)
- ME < 0: systematic **over-prediction** (forecasts too high)
- ME ≈ 0: little average bias

Note: ME can hide large errors due to cancellation, so combine with MAE/RMSE.

Error Metrics II: Scale-Dependent

Mean Absolute Error (MAE):

$$\text{MAE} = \text{mean}(|e_t|) \tag{8}$$

- Linear loss: Each unit of error contributes equally to total error
- Interpretation: Average absolute forecast error in original units

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\text{mean}(e_t^2)} \tag{9}$$

- Quadratic loss: Penalizes large errors more heavily
- Also in original units

When to use: Comparing methods on same series

Error Metrics III: Percentage-Based

Mean Absolute Percentage Error (MAPE):

$$\text{MAPE} = \text{mean}(|p_t|) \quad \text{where} \quad p_t = \frac{100 \times e_t}{y_t}$$

- Unit-free: Compare across series with different scales
- Problem: Undefined if $y_t = 0$; biased if y_t near zero

Further Material:

- More metrics: https://otexts.com/fpp3/accuracy.html
- Metrics for probabilistic forecasts: https://otexts.com/fpp3/distaccuracy.html

Metric Comparison Example: Beer Production

Based on a single train-test split, the seasonal mean method appears to outperform the others. However, this is just one data version and may not be representative. We will see how cross-validation can provide a more robust evaluation.

Metric	Mean	RW	RW+Drift	Seas. Mean
ME	**-8.705**	35.750	32.081	-9.373
RMSE	39.293	52.405	50.396	**13.853**
MAE	35.477	39.750	37.712	**12.079**
MPE	-2.833	7.690	6.810	**-2.190**
MAPE	8.319	8.759	8.315	**2.845**

Table 1: Australian beer production: Seasonal Naïve (MASE < 1) beats other methods

Cross-Validation

The **single train-test split** approach has **limitations**:

- It can be sensitive to how the data is split (e.g., which time period is used for testing).
- It provides only one estimate of test error, which may not be representative of the model's true performance.
- This approach ignores the uncertainty from a single data version $\Rightarrow$ *How representative is the data?*

Cross-validation (CV)

- Apply model(s) to different data versions.
- Acknowledges and quantifies the uncertainty inherent in data preparation.

Time Series Cross-Validation

Challenge: Time series data has temporal dependencies, so we cannot randomly shuffle data for CV.

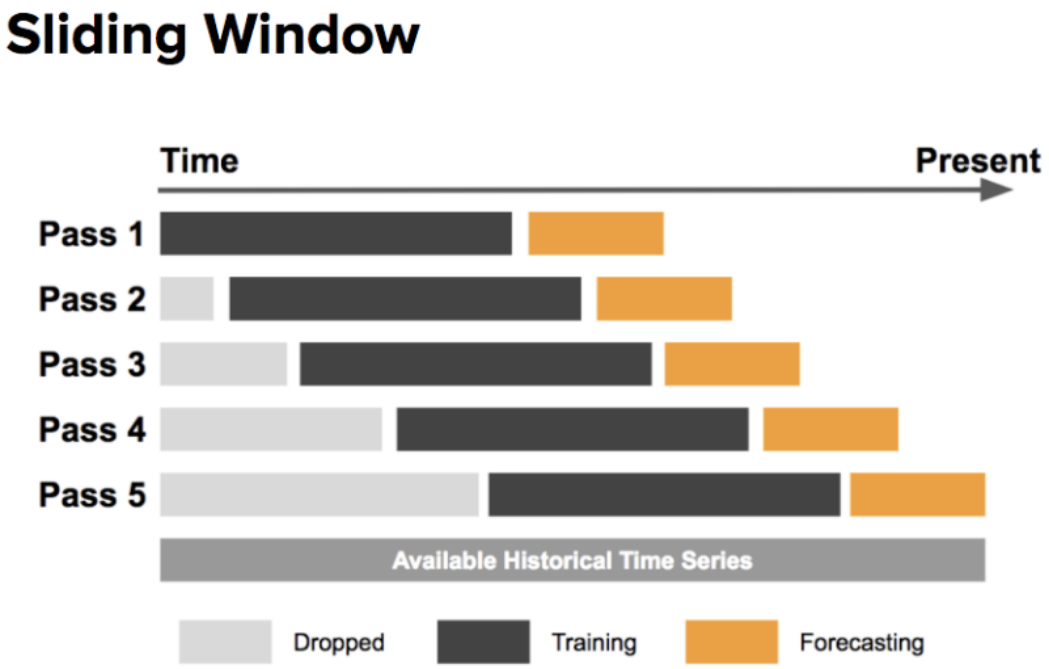


Figure 10: Illustration of rolling (sliding) window CV.

For each pass, we train on a portion of the data and test on the next time period, then slide forward. For obtain multiple h-step forecasts which can

Example for two test-training splits

Stützzeitraum 01.10.2024 – 31.10.2024

Horizon	Date	Actual	Forecast	Error
1	01.11.2024	15	12	3
2	02.11.2024	9	11	-2
3	03.11.2024	21	24	-3
4	04.11.2024	18	19	-1

Stützzeitraum 07.10.2024 – 07.11.2024

Horizon	Date	Actual	Forecast	Error
1	08.11.2024	35	28	7
2	09.11.2024	22	21	1
3	10.11.2024	8	15	-7
4	11.11.2024	11	7	4

Evaluation using all observations

Horizon	Actual	Forecast	Error
1	15	12	3
2	9	11	-2
3	21	24	-3
4	18	19	-1
1	35	28	7
2	22	21	1
3	8	15	-7
4	11	7	4
		ME	0.25
		MdE	0

Evaluation grouped by horizon (only if you have many forecasts)

Horizon	ME
1	5
2	-0.5
3	-5
4	1.5

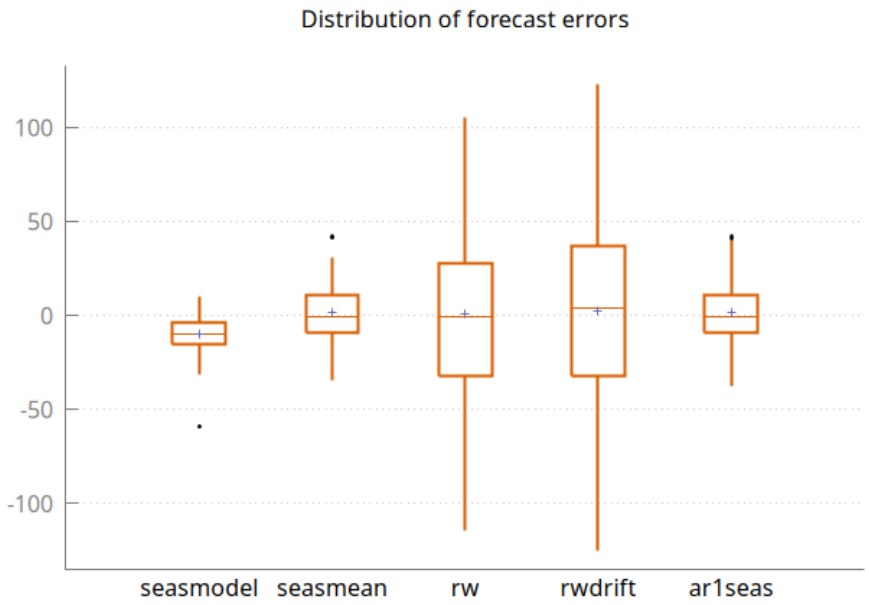
Figure 11: Evaluation of 4-step forecasts using two different test-training splits.

Cross-validation: Example in Beer Production

Key points

- Five models
- Rolling window CV with window size 50 quarters
- $k = 24$ test-training splits (each with 8-step forecasts)
- In total 164 forecasts per model $\Rightarrow$ 164 forecast errors for evaluation

	Seas. model	Seas. mean	RW	RW+drift	AR(1)
ME	-9.52	2.00	0.55	2.18	**1.99**
RMSE	**14.89**	15.22	50.64	54.41	15.70
MAE	11.97	**11.96**	39.76	43.01	12.36
MPE	-2.20	0.37	-0.60	**-0.21**	0.36
MAPE	**2.79**	2.80	9.26	10.02	2.89



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Appendix

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Additional Material and Examples

Forecasting, Goals, and Planning

Forecasts

- Are frequent statistical tasks in business
- Help inform decisions about planning production, transport, and staffing
- Provide guidance for long-term strategic planning
- Business forecasting is often done poorly

What's forecasted?

- **Economics**: GDP growth, inflation, unemployment
- **Finance**: Stock prices, exchange rates, asset returns
- **Operations**: Demand forecasting, inventory management, staffing
- **Planning**: Budget allocation, resource allocation, risk management

Forecasting, Goals, and Planning

Forecast

Seeks to predict the future as accurately as possible, taking into account all available information, including historical data and knowledge of future events that may influence forecasts.

Goals

Goals are what you would like to achieve. Goals should be linked to forecasts and plans, but often are not.
Too often, goals are set without a plan for how to achieve them, and without forecasts of whether they are realistic.

Planning

Is a response to forecasts and goals. Planning involves determining the appropriate actions required to align your forecasts with your goals.

Naive/simple methods as special cases of ARIMA

ARIMA(p, d, q):

- p = AR order (lags of dependent variable)
- d = Differencing (for stationarity)
- q = MA order (lags of errors)

Examples:[4]

- ARIMA(0,0,0) = White noise with no constant
- ARIMA(p,0,0) = AR(p): Autoregressive, no differencing
- ARIMA(0,1,0) = Random walk without drift (no constant)
- ARIMA(0,1,0) = Random walk with drift (with constant)
- ARIMA(0,1,1) = Simple exponential smoothing

Seasonal extension: ARIMA(p,d,q)(P,D,Q)$_s$ for seasonal patterns.

[4] See, e.g., https://otexts.com/fpp3/non-seasonal-arima.html

Cross-Validation Approaches

Expanding Window (preferred):

- Training set grows, test set fixed
- Mimics real-world forecasting: more data accumulates over time
- Better for stable models

Sliding Window:

- Fixed training size slides forward
- Test set moves with training set
- Better if model performance changes over time

Gretl implementation: Manual backtesting loop with forecast generation